SARGap: A Full-Link General Decoupling Automatic Pruning Algorithm for Deep Learning-Based SAR Target Detectors

Jingqian Yu, Jie Chen, *Member, IEEE*, Huiyao Wan, Zheng Zhou, Yice Cao, Zhixiang Huang, *Senior Member, IEEE*, Yingsong Li, *Senior Member, IEEE*, Bocai Wu, and Baidong Yao

***Abstract*— Synthetic aperture radar (SAR) target detectors based on deep learning have difficulty finding a good balance between accuracy and speed. Current pruning methods are usually used for backbone consistent pruning and seldom directly for the whole structure of deep learning target detectors; therefore, for edge-end applications, this article proposes a new full-link general automatic pruning algorithm for SAR target detectors, referred to as SARGap. First, SARGap automatically analyzes the network structure by creating a dependency graph, divides the pair-coupled network structure into the same group, and prunes the same channel for the same group of network structures so that the algorithm can be applied to a variety of complex target detectors. Second, an automatic pruning rate search method (APRS) is designed to search for the optimal pruning rate of each group of network structures in the target detector. Finally, to find a good balance between precision and speed in the automatic search of the pruning rate, a multi-objective optimization loss function (MOOL) is constructed as the APRS objective function. A series of experiments based on SSDD and HRSID, two large-scale SAR target detection datasets, are carried out to prove the superiority of this method. Using Yolov5s as the baseline, SARGap can compress parameters by 84.29%/82.86% and flops by 80.50%/81.93% on two datasets with almost no loss of accuracy. In addition, SARGap can be applied to any deep learning target detector and match hardware computing resources to achieve optimal full-link pruning.**

***Index Terms*— Deep learning, evolutionary algorithm, model compression, synthetic aperture radar (SAR) target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave remote sensing imaging system renowned for its immunity to environmental conditions, including light and weather, and its ability to provide high-resolution imaging day and night. SAR functions by actively transmitting electromagnetic waves toward a target, receiving backscattered signals, and subsequently processing these signals to generate 2-D SAR images. Because of its superior capabilities in target detection when compared to alternative methods, SAR has assumed a pivotal role in various applications, particularly in maritime domains. This includes military and civilian applications such as maritime strategic reconnaissance, marine operations, maritime law enforcement and security, maritime traffic monitoring, and environmental conservation. Central to the extraction of target information and the intelligent interpretation of SAR data is the crucial task of detecting and identifying SAR-equipped vessels [1], [2], [3].

The traditional SAR image target detection algorithm is the constant false alarm rate algorithm, referred to as the CFAR algorithm. The CFAR algorithm is a method to calculate the segmentation threshold based on the SAR image background and target with an unchanged false alarm rate to achieve the best effect [4]. Commonly used CFAR algorithms include global CFAR, two-parameter CFAR, and two-level CFAR [5], [6], [7], [9]. Global CFAR uses the whole SAR image to calculate the segmentation threshold. Because the target and background in the SAR image are very complex, the global CFAR algorithm is relatively efficient, but its accuracy is relatively low. In view of the low accuracy rate of the global CFAR algorithm, the two-parameter CFAR algorithm uses the sliding window method to calculate the segmentation threshold for every pixel in the SAR image. Although this method improves accuracy, it is relatively low in efficiency because of the increased number of computations. Combined with the advantages and disadvantages of the global CFAR algorithm and the two-parameter CFAR algorithm, the two-level CFAR algorithm first uses the global CFAR algorithm to prescreen the target and then calculates the segmentation threshold according to the potential target. Compared with the global CFAR algorithm, the accuracy of the two-level CFAR algorithm has greatly improved. Currently, many traditional SAR image target detection algorithms are improved on the above CFAR algorithms. The CFAR algorithm can often only achieve good performance on the basis of better SAR image imaging results; however, the target and background in SAR images are often very complex, the target is often

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Jingqian Yu, Huiyao Wan, Zheng Zhou, Yice Cao, Zhixiang Huang, and Yingsong Li are with the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui University, Hefei 230601, China (e-mail: q21301178@stu.ahu.edu.cn; huiyaowan1995@163.com; zhouzheng19961123@163.com; yccao@ahu.edu.cn; zxhuang@ahu.edu.cn; liyingsong@ieee.org).

Jie Chen is with the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Anhui University, Hefei 230601, China, and also with the 38th Research Institute of China Electronics Technology Group Corporation, Hefei 230094, China (e-mail: jiechen@ustc.edu).

Bocai Wu and Baidong Yao are with the 38th Research Institute of China Electronics Technology Group Corporation, Hefei 230094, China (e-mail: 18110995593@189.cn; yao1984@mail.ustc.edu.cn).

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classified as background, and the background is classified as the target. Using the CFAR algorithm to detect SAR images, therefore, cannot achieve ideal performance. In recent years, the advantage of SAR target detectors based on deep learning has become increasingly obvious, and the detection performance in many open-source SAR images is significantly better than that detected by traditional methods; therefore, applying SAR target detectors to detect SAR images has become the current development trend [10], [11], [12], [13], [14], [15], [16], [17], [18]. At present, the classical SAR target detector can be divided into a two-stage target detection algorithm and a single-stage target detection algorithm. The classic two-stage target detection algorithms include Faster R-CNN, Mask R-CNN, and RetinaNet. The two-stage target detection algorithm can be divided into three independent running stages: the first stage generates region proposals, the second stage extracts features, and the third stage predicts classification. Compared with the two-stage object detection algorithm, the single-stage object detection algorithm extracts features directly from SAR images and then outputs the target category and target location information. Classic single-stage target detection algorithms include the YOLO series, FPN, and SSD [19]. Since the single-stage detection algorithm is more real time than the two-stage detection algorithm, the single-stage target detection algorithm is often used to detect SAR images in practical engineering applications. Although designing SAR target detectors based on deep learning is very reasonable and SAR image detection accuracy is relatively high, due to the relatively large depth and width of the deep learning detection model, accuracy is given priority; however, the real time and portability are ignored. For example, the classical target detection algorithm Faster R-CNN [20] has a 41.1M model parameter count and 91.4G floating-point operations (FLOPs). The network model FBR-Net [21] is designed for SAR target detection with 32.5M parameters. The current mainstream AI edge deployment devices, however, have low memory capacity and computing power. The Nvidia Jetson Nano device has only 2 GB of storage space and 472G FLOPs per second (FLOPS) computing power. This means that once other basic software and environments are loaded on such an edge device, the remaining available computing resources will not be enough to support the object detection model for the corresponding edge inference application. It is, therefore, necessary to consider some methods to find a good balance between accuracy and speed so that the SAR target detector can be easily deployed in edge devices.

As an important method for compressing the target detector, model pruning removes noncritical redundant parameters to reduce the number of computations and parameters in the deep learning detection model and to ensure that the detector can maintain good performance after pruning parameters. Traditional pruning methods can be divided into structured pruning [22], [23], [24] and unstructured pruning [25], [26], [27]. Compared with unstructured pruning, structured pruning does not require specific underlying hardware or computational acceleration library to achieve the expected effect, so structured pruning has been widely used in practice [28], [29]. At present, there are intricate connections in the basic modules, such as the convolutional layer and batch normalization layer in the target detector. When channel pruning is performed on a certain network structure, all its dependent layers need to be pruned accordingly; otherwise, the detection model obtained will be broken [30], [31]. The current structure of the target detector network is, however, infinitely complex [31], [32], so the current pruning methods usually have a difficult problem; that is, the algorithm implementation and the network model are strongly bound. In addition, current pruning methods are usually for backbone consistent pruning. If these pruning methods are replicated into the whole structure of a deep learning target detector, it is necessary to manually analyze the dependency relationships among the target detectors, which will waste considerable energy. In addition, even if the algorithm is implemented, the dependencies in the detector need to be reanalyzed, and the algorithm needs to be reimplemented in the face of a new target detector [23], [33]. Fang et al. [34] proposed automating the dependency relationship analysis in classification network models by creating dependency graphs, which undoubtedly saved much energy compared with manually analyzing dependency relationships in network models; however, this method is mainly aimed at consistent backbone pruning in natural scenes and does not consider the automatic pruning design of end-to-end target detectors.

For SAR target detection tasks, however, there is no end-to-end automatic pruning method for deep learning target detectors.

In light of these considerations, this article introduces an inventive, comprehensive automatic pruning algorithm tailored for SAR target detectors in edge-end applications, named SARGap. SARGap is specifically designed to cater to a broad spectrum of deep learning-based SAR target detectors. It operates autonomously, taking into account the computational resource constraints of edge computing devices. Its primary objectives are to strike a harmonious equilibrium between target detection accuracy and computational efficiency while facilitating end-to-end automatic pruning of the target detector without requiring human intervention.

The main contributions of this article can be summarized as follows.

1) A general structured pruning strategy for arbitrary SAR target detectors called the GSP algorithm, is designed. This strategy can automatically group all the network structures of any target detector. First, GSP automatically analyzes the interlayer and intralayer dependencies of the network structure. The dependency graph is then created based on the dependency relationships. Finally, according to the dependency graph, the pairwise coupled network structures are automatically divided into groups. Pruning the same channels for the same group of network structures prevents the target detector from becoming a broken detector after pruning.
2) Theparameter redundancy of different network structures is also different; manually setting the pruning rate is too subjective, and the efficiency is limited. To automatically search for different group pruning rates, an automatic pruning rate search method called the “APRS algorithm”

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is designed. The algorithm can automatically search for the optimal pruning rate combination without manual participation.

3) A new multiobjective optimization loss function called "MOOL" is constructed, which is composed of precision and a number of calculations and parameters. It is used to guide the model to automatically search for the optimal pruning rate combination that can balance accuracy and speed according to the computing resources and precision index requirements of the hardware platform.
4) The proposed method can be used as a general end-to-end automatic pruning framework for SAR target detectors and is suitable for any deep learning target detection network.

II. Related Work

At present, the traditional CFAR algorithm can only achieve good performance on the basis of better SAR image imaging results. In many SAR target detection datasets, the performance of SAR target detection based on deep learning is significantly better than that of traditional SAR target detection methods; therefore, end-to-end SAR target detectors based on deep learning are increasingly widely used in SAR target detection; however, the end-to-end SAR target detector based on deep learning is not well balanced among accuracy and a number of parameters and computations. Considering the computing resource tolerance of edge devices, reducing the number of parameters and computations while retaining the detection model performance has become the focus of research. In recent years, many researchers have developed lightweight SAR target detectors specifically for SAR images and pruned the backbone of the target detector by the model pruning method to reduce the number of parameters and computations.

A. SAR Target Detection Method Based on Deep Learning

SAR target detection methods based on deep learning can be divided into anchor-based detection methods and anchor-free detection methods according to the network structure. The key difference between the two detection methods is whether the predefined anchor boxes are used to match the real target box. Anchor-based detection methods use anchor boxes of different shapes and sizes in advance to regress and classify targets. This detection method is usually modeled as a problem of classifying and regressing some candidate regions. Representative detection methods based on anchor-based methods include Faster R-CNN, SSD, and YOLO series [19]. Detection methods based on anchor-free methods can be divided into two target detection algorithms based on the central domain and key point. The target detection method based on the central domain predicts the central coordinates of the object and the scale information of the boundary box directly. For the keypoint-based target detection algorithm, the probability that each point in the input image is the key point in the boundary box is predicted by a thermal map, and then multiple groups of thermal maps are combined to obtain the object boundary box. Representative anchor-free target detection algorithms include YOLOX, CenterNet, CornerNet, and FCOS [19]. Both anchor-based and anchor-free algorithms rely on a deep network model, and there is no good balance between precision and numbers of parameters and computations.

B. Lightweight SAR Target Detector

Chang et al. [35] took YOLOV2 as the basic framework to seek a balance between accuracy and speed and designed a lightweight SAR target detector with fewer layers. Later, Zhou et al. [36] designed a lightweight feature extractor with YOLOV3 as the basic framework and built a new feature aggregation module based on transformer architecture, obtaining a lightweight SAR image small sample target detection method. Recently, Wan et al. [17] redesigned the lightweight backbone and attention enhancement PAN module using YOLOX as the basic framework and proposed a new target detection head that can reduce the number of parameters while maintaining accuracy. Feng et al. [37] also used YOLOX as a basic framework to balance accuracy and speed through a redesigned lightweight multiscale backbone and a new position-enhanced attention strategy. Meanwhile, Zhou et al. [38] designed a hybrid representation learning-enhanced SAR target detection algorithm with YOLOV5 as the basic framework, reducing the number of parameters and computations and ensuring good performance. Although these methods have made some contributions to the lightweight SAR target detection algorithms, implementing these algorithms is for a single network model. When facing a new network model, according to t edge device computing resources, it is necessary to manually participate in the lightweight processing of the network model.

C. Pruning Method of the Target Detection Model

Model pruning reduces the number of computations and parameters in a deep learning model by removing noncritical redundant parameters. Unstructured pruning is based on convolution kernel internal pruning, which can be divided into fine-grained pruning and vector-level pruning. Fine-grained pruning is the smallest grain of pruning, directly pruning the connections and neurons. Vector-level pruning is only slightly larger than fine-grain pruning. Unstructured pruning can make the model achieve good compression while maintaining good model detection performance; however, the sparse structure of the model after unstructured pruning can achieve good performance improvement only when there is a special underlying hardware or a computational acceleration library supporting its sparse structure [39]. Structured pruning prunes the number of convolutional cores and channels and can be divided into kernel-level pruning, filter-level pruning, and layer pruning. Kernel-level pruning is the convolution operation corresponding to reducing the input channels. Filter-level pruning is the operation of removing the convolution kernel set corresponding to the output channel. Layer pruning is the removal of an entire layer in the network, which causes damage to the network structure that is difficult to recover, so the layer pruning method is generally not used. Kernel-level pruning and filter-level pruning can easily change the model structure, and good

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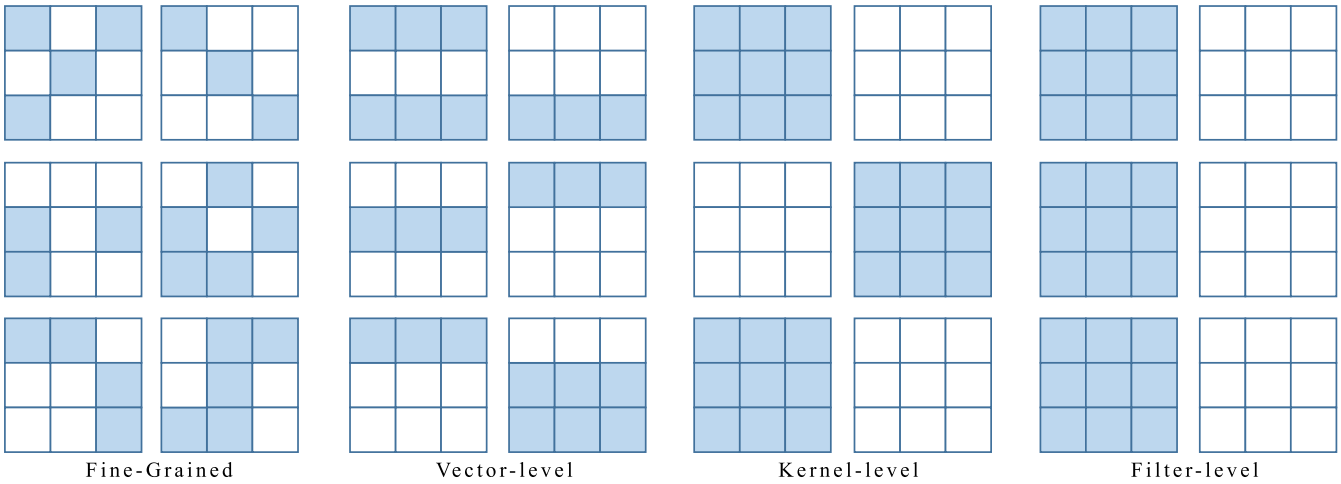


Fig. 1. Unstructured pruning and structured pruning diagrams. The blue section shows the pruning parameters.

performance can be achieved without special underlying hardware or computational acceleration libraries, so kernel-level pruning and filter-level pruning are more commonly used than unstructured pruning and layer pruning. Fig. 1 shows the pruning methods of fine-grained pruning, vector-level pruning, kernel-level pruning, and filter-level pruning, respectively.

Since the sensitivity and redundancy of different network structure layers in the deep learning detection model are different, if the pruning rate of different network structure layers is manually set, considering the complexity and size of the network model, it will waste considerable time, and it is difficult to achieve the optimal solution. Currently, many automated pruning methods have emerged to explore the optimal pruning rate of different network structure layers of the backbone in object detection models [40], [41], [42], [43], [44], [45], [46]. For example, He et al. [41] proposed a method based on reinforcement learning to explore network model compression strategies and designed a reinforcement learning agent, the deep deterministic policy gradient (DDPG), to output the corresponding pruning rate of each layer according to state information such as parameter number and minimum pruning rate. Experiments show that this reinforcement learning-based pruning strategy is superior to the traditional artificial design model compression strategy. There is, however, a problem that cannot be ignored in these automatic pruning methods; that is, the algorithm implementation and the network structure have strong binding; moreover, all the existing structured pruning algorithms are implemented on the backbone of the object detection model. The automatic pruning design for end-to-end target detectors was not considered.

A general full-link automatic pruning algorithm for arbitrary SAR target detectors is, therefore, designed in this article.

III. Methods

SARGap automates the analysis of network structures within SAR target detectors, grouping them intelligently. To optimize precision and speed, we employ the MOOL function as the objective for the APRS method. This enables automatic exploration of pruning rates for each network structure group; furthermore, SARGap facilitates the deployment of complex target detectors on resource-constrained edge devices, preserving excellent performance despite limited computational resources and storage capacity.

A. General Introduction of the SARGap Compressed SAR Target Detector

As shown in Fig. 2, SARGap first automates the analysis of the overall network structure of the SAR target detector and then automatically groups all the network structures according to the analysis results. Then, SARGap divides those pairwise coupled network structures into a group and prunes the same channel for the same group of network structures to avoid the target detector becoming a broken detector after pruning. Second, since the sensitivity and redundancy of each layer of the network structure in the SAR target detector are different, this article uses the APRS method to automatically search for the pruning rate of each group of network structures. To find a good balance between precision and speed when automatically exploring the optimal pruning rate of each group of network structures, the MOOL function is created as the objective function of APRS. The MOOL function is a function that can be supervised for precision and number of computations and parameters. Finally, a set of pruning rates is found through the APRS algorithm, which makes the optimal MOOL function value meet the restriction condition. Next, this article introduces how to automatically analyze the network structure of the target detector and divide the pair-coupled network structure into the same group, how the APRS algorithm automatically searches for the best pruning rate, and how the MOOL function supervises the precision and number of parameters and computations simultaneously.

B. Target Detector Network Structure Grouping

Structured pruning can be divided into input channel pruning and output channel pruning, so to facilitate the dependency description, this article divides each layer L_i into input layer L_i^- and output layer L_i^+. If the target detector has a

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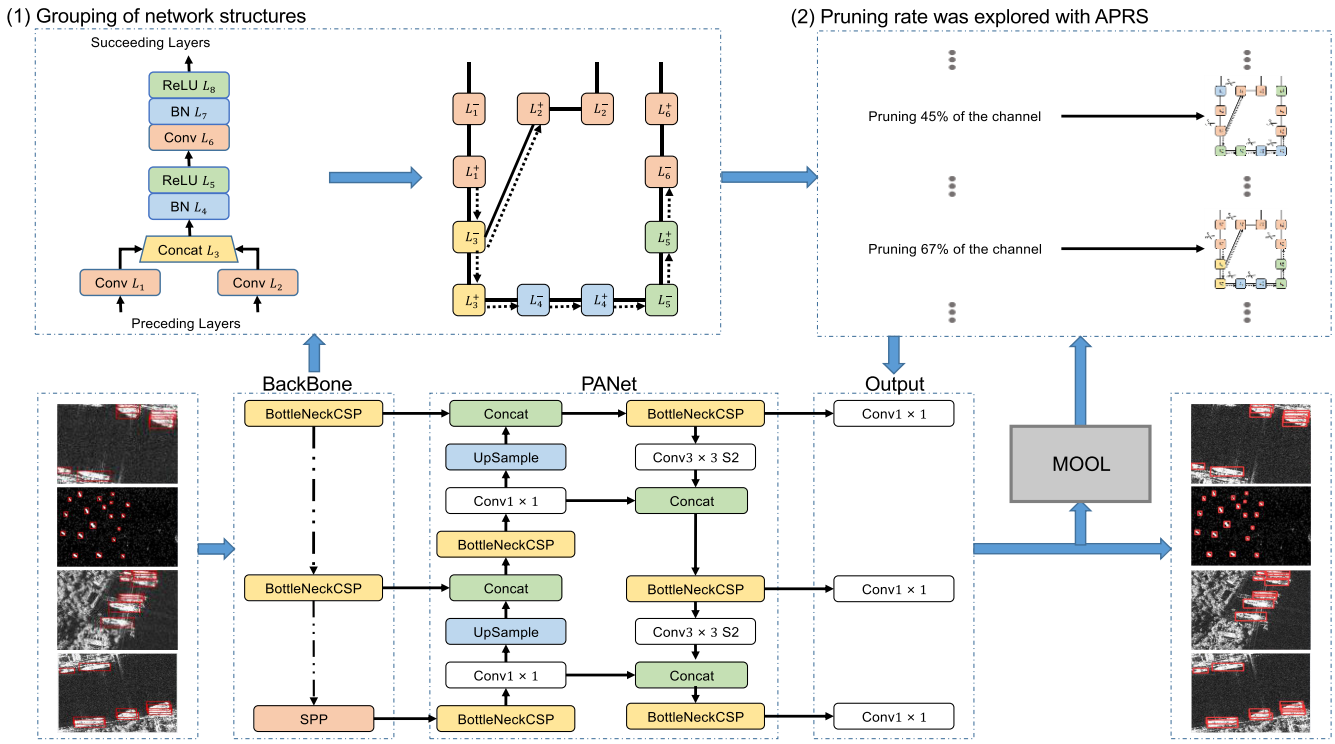


Fig. 2. SARGap overall compression process diagram.

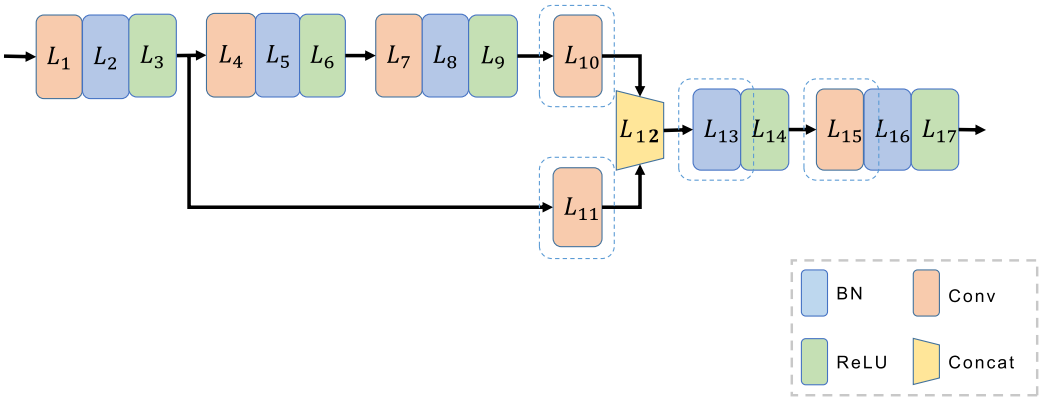


Fig. 3. Csp2_1 structure diagram of yolov5, with the network structure to be pruned inside the blue dashed box.

l-layer network structure, the input layer in the model can be described as $L = \{L_1^-, L_2^-, \ldots, L_l^-\}$, and the output layer can be described as $L = \{L_1^+, L_2^+, \ldots, L_l^+\}$. There may be dependencies between different basic modules in a target detector with a complex internal structure, and the dependent network structure must be pruned simultaneously when a layer is structured. As shown in Fig. 3, when pruning the output layer L_{10}^+ of convolutional layer L_{10}, corresponding channels must be pruned for input layer L_{13}^- and output layer L_{13}^+ of L_{13}, output layer L_{11}^+ of L_{11} and input layer L_{15}^- of L_{15}; otherwise, the target detector after pruning will be a broken target detector. In this article, the dependencies between network structures are divided into interlayer dependencies and intralayer dependencies.

1) Interlayer Dependencies: Interlayer dependencies are those that result from direct connections between layers. The output layer L_i^+ of the upper layer and the input layer L_j^- of the next layer correspond to the same feature map. During channel pruning, "output channel pruning of the upper layer" and "input channel pruning of the next layer" are equivalent, so the two layers need to prune the same channel. In this article, the dependence between layers is described as $L_i^+ \leftrightarrow L_j^-$.

2) Intralayer Dependencies: Intralayer dependencies are closely related to the nature of the layer. The input and output of some basic modules in the target detector are independent and uncoupled. For example, the input and output of the convolutional layer can be independently pruned, and each can have a different pruning layout; thus, the convolutional layer does not have an intralayer dependency relationship. This nonexistent intralayer dependency is described as $\text{sch}(L_i^-) \neq \text{sch}(L_i^+)$. There are also some basic modules

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whose inputs and outputs are coupled, such as the batch normalization layer, whose inputs and outputs have the same pruning layout, which affects the number of input and output channels simultaneously; therefore, batch normalization layers have intralayer dependencies. The target detector usually has a very complex network structure, but its layer types are limited, so the intralayer dependencies of different layers can be defined in advance. This article describes the intralayer dependencies as $\text{sch}(L_i^-) = \text{sch}(L_i^+)$.

The key to automatically grouping network structures is automatically analyzing the dependencies of complex structures in the target detector. Although it is feasible to manually analyze dependencies in detectors today, it often requires considerable effort when facing complex detectors, and the algorithm specifically designed according to manual analysis dependencies cannot be used for a new target detector; therefore, to solve this problem, this article creates a binary dependency graph $D \in \mathrm{R}^{2l\times 2l}$, where $D_{ks} = 1$ indicates that there is a dependency relationship between layers k and s, where $k \in \{1, 2, \ldots, 2l\}$, $s \in \{1, 2, \ldots, 2l\}$.

The dependency model is established according to the above two dependency relationships

$$\begin{aligned} &D\left(L_j^-, L_i^+\right) \\ &\quad = D\left(L_i^+, L_j^-\right) \\ &\quad = 1\left[L_i^+ \leftrightarrow L_j^-\right] \vee 1\left[i = j \wedge \text{sch}\left(L_i^-\right) = \text{sch}\left(L_j^+\right)\right] \end{aligned} \tag{1}$$

where $i \in \{1, 2, \ldots, l\}$, $j \in \{1, 2, \ldots, l\}$, 1 is an indicator function that returns the condition "true," $\vee$ represents the "OR" operation, and $\wedge$ represents the "AND" operation. $1[L_i^+ \leftrightarrow L_j^-]$ summarizes the interlayer dependencies, and $1[i = j \wedge \text{sch}(L_i^-) = \text{sch}(L_j^+)]$ summarizes the intralayer dependencies.

The dependency between adjacent layers can be recursive. For example, if there is a dependency between adjacent layers A and B and a dependency between adjacent layers B and C, then there is still a dependency between A and C, although they are not directly connected; therefore, grouping by dependency graph can be seen as a path search problem. As shown in Fig. 4, taking L_{10}^+ as the starting point, there is a dependency relationship between L_{10}^+ and L_{12}^- because $D(L_{10}^+, L_{12}^-) = 1$. Then, from the starting point of L_{12}^-, we know $D(L_{11}^+, L_{12}^-) = 1$ and $D(L_{12}^+, L_{12}^-) = 1$, and therefore, there is a dependency between L_{11}^+ and L_{12}^- and L_{12}^+ and L_{12}^-. Then, there is also a dependency relationship between L_{10}^+, L_{11}^+, and L_{12}^+. Then, recursion is performed step by step. Finally, it can be understood that L_{10}^+, L_{12}^-, L_{11}^+, L_{12}^+, L_{13}^-, L_{13}^+, L_{14}^-, L_{14}^+, and L_{15}^- all have a dependency relationship with each other; therefore, when the L_{10}^+ channel is pruned, L_{11}^+, L_{13}^-, L_{13}^+, and L_{15}^- also need to be pruned accordingly.

C. Multiobjective Optimization Loss Function: MOOL

The current loss function used by most target detectors is the binary cross-entropy loss (BCELoss) function. Directly applying the BCELoss function as the objective function can only contribute to the target detector accuracy when automatically searching the pruning rate of the SAR target detector; however, it has no contribution to either the number of computations or parameters. This article, therefore, designs the multiobjective optimization loss function (L_{MOOL})

$$L_{\text{MOOL}} = L_{\text{BCEL}} + \lambda_{\text{flops}}\log_e\text{flops} + \lambda_{\text{params}}\log_e\text{params} \tag{2}$$

where L_{BCEL} is the BCELoss function. Assuming p_i is the prediction probability of the target class and y_i is the target class label, the formula can be expressed as follows:

$$L_{\text{BCEL}} = \frac{1}{N}\sum_{i=1}^{N} -\left[y_i \cdot \log(p_i) + (1 - y_i) \cdot \log(1 - p_i)\right]. \tag{3}$$

L_{BCEL} can ensure the target detector's accuracy. flops and params are the numbers of target detector computations and parameters, respectively, and $\lambda_{\text{flops}}\log_e\text{flops}$ and $\lambda_{\text{params}}\log_e\text{params}$ can ensure that the number of target detector computations and parameters are limited within a certain range. λ_{flops} and λ_{params} are the balance factors of the computations and the number of parameters, respectively; therefore, L_{MOOL} is a function that contributes to precision and a number of computations and parameters. Using L_{MOOL} as the objective function to automatically search the target detector pruning rate can, therefore, ensure the detector accuracy and limit the number of calculations and parameters.

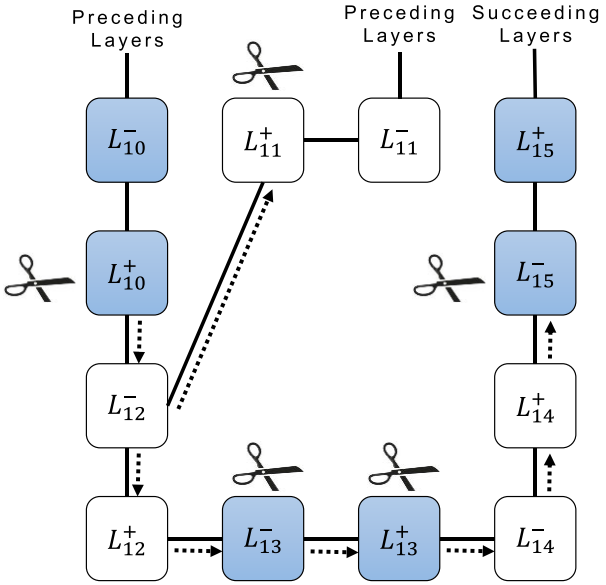


Fig. 4. As the starting point of, the grouping mode is realized by a dependency graph.

D. Exploring the Pruning Rate by the APRS Algorithm

APRS is an automatic pruning rate search algorithm. The APRS algorithm first randomly initializes the population containing NP particles. If the target detector has a total D group network structure that needs to be pruned, each particle is set as a D-dimensional vector

$$X_i = (x_{i1}, x_{i2}, \ldots, x_{iD}) \tag{4}$$

where $i \in \{1, 2, \ldots, NP\}$. The population is then randomly divided into Group P subpopulations, each containing

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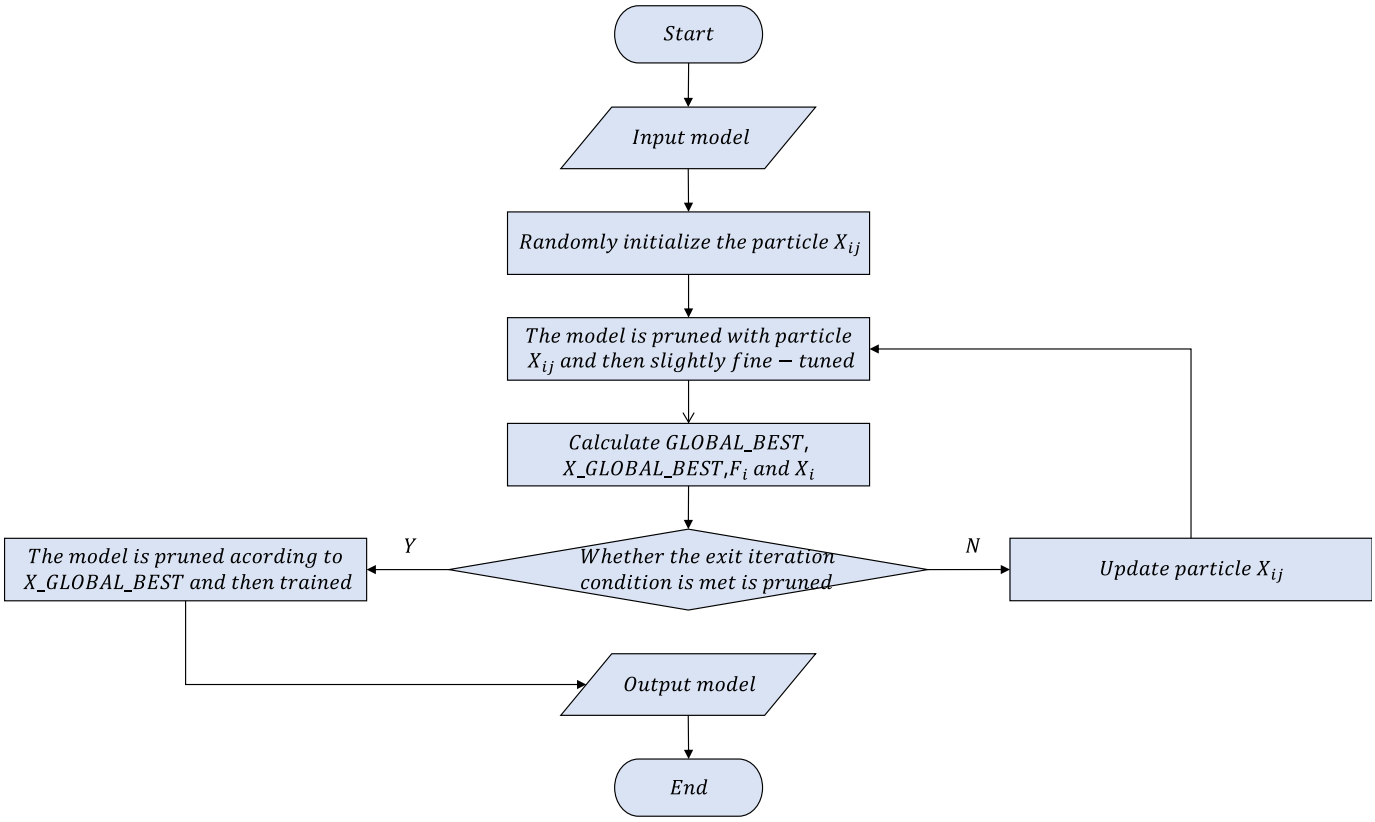


Fig. 5. Specific search process of the SARGap compressed target detector.

N particles. Then, each particle can be represented as

$$X_{ij} = (x_{ij1}, x_{ij2}, \ldots, x_{ijD}) \tag{5}$$

where $i \in \{1, 2, \ldots, P\}$, $j \in \{1, 2, \ldots, N\}$. Assuming that the objective function is $f(x) = L_{\text{MOOL}}$, find the objective function value for all the particles in the subpopulation

$$F_{ij} = f(X_{ij}) \tag{6}$$

where $i \in \{1, 2, \ldots, P\}$, $j \in \{1, 2, \ldots, N\}$. Then, according to the obtained objective function value, determine the optimal objective function value of all the objective function values

$$F_\text{GLOBAL_BEST} = \min(F_{ij}) \tag{7}$$

where $i \in \{1, 2, \ldots, P\}$, $j \in \{1, 2, \ldots, N\}$. The corresponding particle X_GLOBAL_BEST, where

$$F_\text{GLOBAL_BEST} = f(X_\text{GLOBAL_BEST}). \tag{8}$$

In addition, find the local optimal function values F_LOCAL_BEST for all subpopulations

$$F_i = \min(F_{i1}, F_{i2}, \ldots, F_{iN}) \tag{9}$$

$$F_\text{LOCAL_BEST} = (F_1, F_2, \ldots, F_P). \tag{10}$$

The corresponding particle is found according to the local optimal function value of each subpopulation

$$F_i = f(X_i) \tag{11}$$

$$X_\text{LOCAL_BEST} = (X_1, X_2, \ldots, X_P). \tag{12}$$

In each iteration, the APRS algorithm updates the jth particle x_{ij} in subpopulation i by tracking the particle X_GLOBAL_BEST corresponding to the optimal objective function value F_GLOBAL_BEST and the particle X_i corresponding to the local optimal function value F_i in its subpopulation

$$v_{ij} = w \times v_{ij} + c_1 r_1 (X_i - X_{ij}) + c_2 r_2 (X_\text{GLOBAL_BEST} - X_{ij}) \tag{13}$$

$$X_{ij} = X_{ij} + v_{ij} \tag{14}$$

where v_{ij} represents the update speed and w is the inertia weight, representing the ability to inherit the previous speed. The larger w is, the stronger its global search ability. In contrast, the local search ability is stronger. c_1, c_2 are learning factors, and r_1, r_2 are random numbers in the range [0, 1].

E. SARGap Overall Search Process

As shown in Fig. 5, the specific search process of the SARGap compressed target detector is as follows.

1) In the first APRS iteration, a series of particles X_{ij} is randomly initialized, and each particle X_{ij} is a set of numbers between $[x_{\min}, x_{\max}]$ and length D, where $x_{\min}$ represents the lower limit of the pruning rate and $x_{\max}$ represents the upper limit of the pruning rate. Each number in particle X_{ij} represents the pruning rate for

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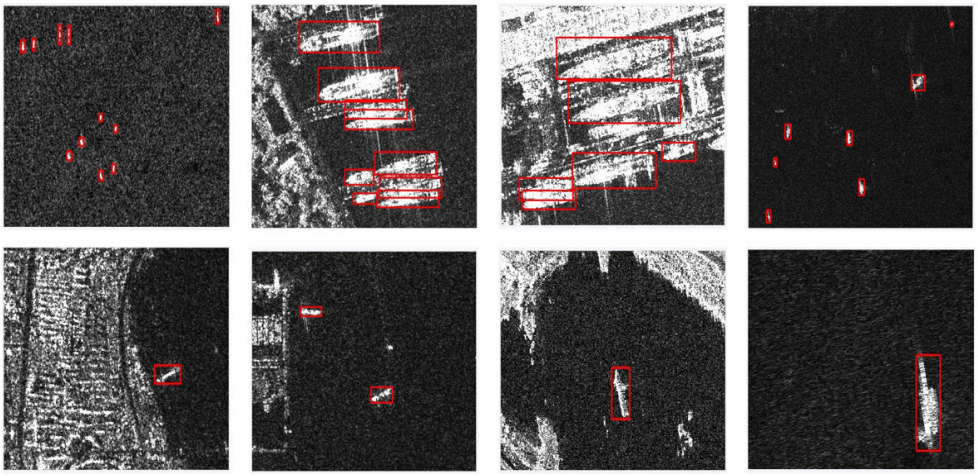


Fig. 6. Sample image on the SSDD dataset with red boxes representing real targets.

each set of network structures to be pruned in the target detector.

2) Gradually pass each particle X_{ij} into the target detector for pruning, slightly fine-tune the model detector after pruning, and then use function L_{MOOL} to calculate the objective function value F_{ij} in the APRS algorithm.
3) Then, find the global optimal function value F_GLOBAL_BEST and local optimal function value F_LOCAL_BEST and their corresponding particles X_GLOBAL_BEST and X_LOCAL_BEST according to the obtained objective function value F_{ij}.
4) Observe whether the global optimal function F_GLOBAL_BEST meets the conditions. If it does not meet the conditions, update each particle X_{ij} by particle X_GLOBAL_BEST and particle X_i corresponding to the local optimal function value F_i of its own subpopulation. Then, proceed to step 2). If yes, X_GLOBAL_BEST is regarded as the globally optimal particle and passed into the target detector for pruning. Finally, the pruned model is trained to obtain the final target detector and exit the iterative process.

IV. Experiments

In this article, the SSDD dataset and HRSID dataset were used to compress SARGap in the YOLOV3-tiny, YOLOV5, and YOLOX-tiny detection models to prove the feasibility of SARGap. This section initiates with an introductory overview of the SSDD and HRSID datasets, accompanied by a delineation of the evaluation metrics employed in our experimentation. We then provide a rationale for the necessity of conducting a fine-tuning process spanning ten epochs during each pruning iteration. Following this, we proceed to execute the pruning process on the state-of-the-art target detector using the SARGap method.

A. Dataset Information and Experiment Setup

To verify the feasibility of the SARGap compressed target detector in various complex SAR images, this article conducted validation on the SSDD dataset and HRSID dataset. Next, the SSDD dataset, HRSID dataset, and the evaluation indicators used will be introduced.

1) SSDD is a publicly available dataset for SAR ship target detection and is the first publicly available dataset in its field. There are 1160 SAR images in the SSDD dataset, including 2456 ship targets. The average number of ship targets per image is 2.12, and most of the ship targets are very small. Although the number of images in the SSDD dataset is relatively small, since the SSDD dataset contains only one category of ships, the dataset is sufficient for training models. To date, nearly half of the SAR target detector papers based on deep learning have used the SSDD dataset, which shows the considerable influence of the SSDD dataset in SAR image detection. Fig. 6 shows sample images on the SSDD dataset. There are several reasons why SSDD has been so successful.
   a) First, it is noteworthy that the SSDD dataset predates the second dataset in SAR image detection by approximately one and a half years. Consequently, many algorithmic approaches in research rely primarily on the SSDD dataset due to its early availability.
   b) Second, the SSDD dataset exhibits a rich diversity of sample images originating from different sensors, polarizations, resolutions, sea states, and ship scenes; furthermore, it encompasses challenging samples that demand rigorous attention; therefore, models trained using the SSDD dataset are renowned for their reliability and stability.
   c) Third, the pervasive use of the SSDD dataset in early research endeavors has resulted in subsequent researchers also adopting it. This adoption facilitates direct comparisons between their methodologies and those of their predecessors. Consequently, despite the existence of newer datasets in the realm of SAR image detection, the SSDD dataset maintains its wide adoption.
   d) Fourth, it is imperative to recognize that early SAR image detection research was constrained by limited computational GPU resources. In this context, the SSDD dataset's sample size proves advanta-

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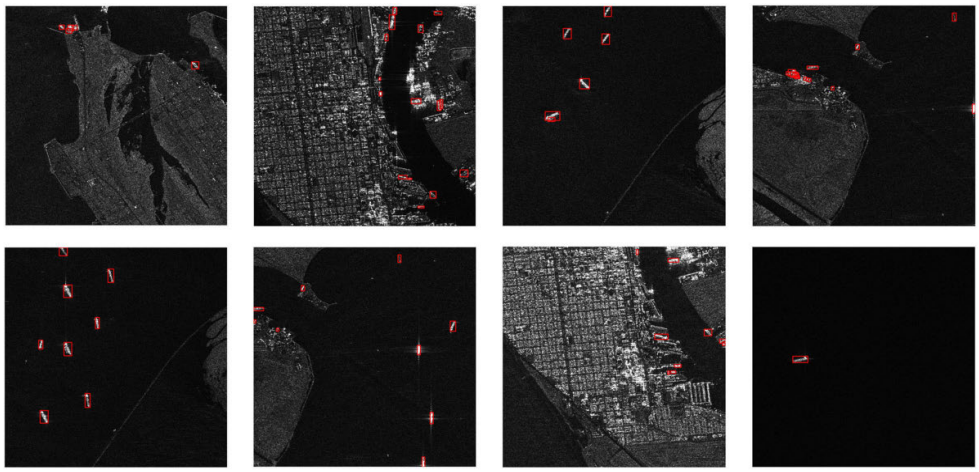


Fig. 7. Sample images on the HRSID dataset, with red boxes representing real targets.

geous for research and development when compared to large-scale datasets within the SAR imaging domain. Additionally, the moderate sample count within the SSDD dataset streamlines model debugging and enhances overall work efficiency.

2) HRSID is a high-resolution dataset published by the University of Electronic Science and Technology of China, which contains a total of 5604 SAR images and 16 591 ship targets. HRSID is a dataset that can be used for semantic segmentation, ship detection, and instance segmentation tasks. The HRSID dataset contains SAR image samples of different resolutions, different sea areas, and different coastal ports. To verify the effectiveness of the model automatic pruning method in different scenarios, this article also uses the HRSID dataset for experiments. Fig. 7 shows sample images on the HRSID dataset.
3) As shown in Table I, to facilitate comparison with previously implemented lightweight network models, this article uses the same training parameters for the SSDD and HRSID datasets as previous researchers. When training the SSDD dataset, the picture size is uniformly set to 640 × 640, where 80% is set as training data, and the remaining 20% is set as test data. When training the HRSID dataset, we set the image size to 800 × 800, set 65% as training data, and the remaining 35% as test data. The batch value of the two datasets is set to 16, and the maximum pruning rate and the minimum pruning rate on the SSDD dataset are set to 80% and 20%, respectively, when the pruning rate of the detector is automatically searched. The maximum pruning rate on the HRSID dataset is set to 80%, and the minimum pruning rate is set to 30%. Each pruned model is slightly fine-tuned ten epochs during the search process. The experiments in this article were performed on the NVIDIA GeForce RTX 3080 super and Ubuntu 18.04 operating systems.

B. Evaluation Index

In this article, mAP is used as the final detection performance index of the detector. mAP is an abbreviation of mean average precision, which is an important indicator to measure the performance of target detection algorithms. The formula for mAP can, therefore, be expressed as

$$\text{mAP} = \text{sum of average accuracy of all classes divided by the number of all classes.}$$

It is assumed that there are two classification target types: positive examples and negative examples. Before calculating mAP, we first introduce a few model evaluation terms.

1) *True Positives (TP):* The number of correctly predicted positives; that is, the number of samples that are actually positive and predicted by the classifier to be positive.
2) *False-Positives (FP):* The number of incorrectly predicted positives; that is, the number of samples that are actually negative and predicted by the classifier to be positive.
3) *False-Negatives (FN):* The number of incorrectly predicted negative cases; that is, the number of samples that are actually positive and predicted by the classifier as negative cases.
4) *True Negatives (TN):* The number of correctly predicted negative cases; that is, the number of samples that are actually negative and predicted by the classifier as negative cases.
5) Precision (P) is the proportion of the positive samples contained in the correct prediction sample to all positive samples in the prediction sample. The formula for P can, therefore, be expressed as

$$\text{P} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{15}$$

6) Recall (R) is the proportion of the positive samples in the correctly predicted sample to all the actual positive samples. The formula for R can, therefore, be expressed as

$$\text{R} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{16}$$

In general, the higher the recall (R) is, the lower the precision (P).

The mAP is based on the area of the precision (P) and recall (R) curve, which is an integral area of PR. The formula

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TABLE I
DATASET INFORMATION AND EXPERIMENTAL SETTINGS

	SSDD	HRSID
images size for training	640×640	800×800
train:valid	8: 2	6.5: 3.5
number of class	1	1
class	ship	ship
Batch size	16	16
fine-tuning epochs for automated exploration	10	10
x_{min} and x_{max} for automated exploration	[0.20,0.80]	[030,0.80]

TABLE II
COMPARISON EXPERIMENT BETWEEN THE FINE-TUNING RESULTS AND THE FINAL TRAINING RESULTS ON THE SSDD DATASET

Method	Params(M)	FLOPs(G)	fine tuning mAP50(%)	final result mAP50(%)
20%(YOLOV5s)	0.3	0.8	84.2	97.4
40%(YOLOV5s)	1.1	2.8	91.8	98.2
60%(YOLOV5s)	2.5	5.9	96.9	98.5
80%(YOLOV5s)	4.5	10.2	97.8	98.8
base(YOLOV5s)	7.0	15.8	98.3	98.9

TABLE III
COMPARISON EXPERIMENT BETWEEN THE FINE-TUNING RESULTS AND THE FINAL TRAINING RESULTS ON THE HRSID DATASET

Method	Params(M)	FLOPs(G)	fine tuning mAP50(%)	final result mAP50(%)
20%(YOLOV5s)	0.3	1.2	74.9	90.7
40%(YOLOV5s)	1.1	4.4	84.7	92.6
60%(YOLOV5s)	2.5	9.3	91.1	93.7
80%(YOLOV5s)	4.5	16.2	92.4	94.1
base(YOLOV5s)	7.0	24.9	94.0	94.5

for mAP can, therefore, be expressed as

$$\text{mAP} = \int_0^1 P(\text{R})\text{dR}. \tag{17}$$

To meet the needs of practical applications, this article also uses the number of parameters (Params) and floating point operations (FLOPs) as evaluation indices.

1) The number of parameters (Params), which indicates how many parameters are contained in the target detector, independent of the input data, is mainly determined by the detector network structure. The number of parameters directly affects the memory or video memory size of the detector.
2) Floating point operations (FLOPs) are used to measure the algorithm complexity. This article uses GFLOPs to represent it, where 1 GFLOPs $= 10^9$ FLOPs.

C. Comparison Experiment Between the Fine-Tuning Results and the Final Training Results Under Different Pruning Rates

The traditional pruning method is oriented to the consistent pruning of the backbone, the network structure is relatively simple, and the algorithm implementation is strongly tied to the detection model. Unlike traditional pruning methods, SARGap faces a more complex end-to-end target detector; however, to overcome the strong binding problem, SARGap automatically analyzes the network structure and divides the pairwise coupling network structure into the same group. The parameter redundancy of network structures in the same group is not the same. If a network structure is randomly selected from a group of network structures for parameter importance ranking, then the unimportant parameters are pruned according to the ranking. Then, the FPBCE function value of the detector is calculated. The experimental results are not ideal. The accuracy obtained by the detector at this time cannot effectively express the advantages and disadvantages of the detector's performance. To solve this problem, SARGap fine-tuned the detector ten epochs after each pruning to see how well the model performed at that time. As shown in Tables II and III, 20%, 40%, 60%, and 80% of the channels of the YOLOV5s detector are pruned. By comparing the detector accuracy after fine-tuning ten epochs and the accuracy obtained by the final training, it is obvious that the detector accuracy after fine-tuning ten epochs on both the SSDD dataset and HRSID dataset can well express the detection performance of the detector at this time. For example, on the SSDD dataset, the performance of the model pruned with a 40% channel number is better than that of the model pruned with a 20% channel number after fine-tuning ten epochs. Additionally, the final training result of the model pruning with a 40% channel number is better than the final training result of the model

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TABLE IV
RESULTS OF EXPERIMENTS CONDUCTED ON THE SSDD DATASET

	policy	Params(M)	FLOPs(G)	mAP50(%)
YOLOV3-tiny[47]	Base	8.7	13.0	98.6
	40% Pruned	3.1	4.8	98.5
	Ours	**1.1**	**2.2**	98.3
YOLOV5s[48]	Base	7.0	15.9	98.9
	Pruned40%	2.5	5.9	98.5
	Ours	**1.1**	**3.1**	98.7
YOLOX-tiny[49]	Base	5.0	15.1	96.2
	40% Pruned	1.8	5.6	96.0
	Ours	**1.1**	**3.7**	96.0

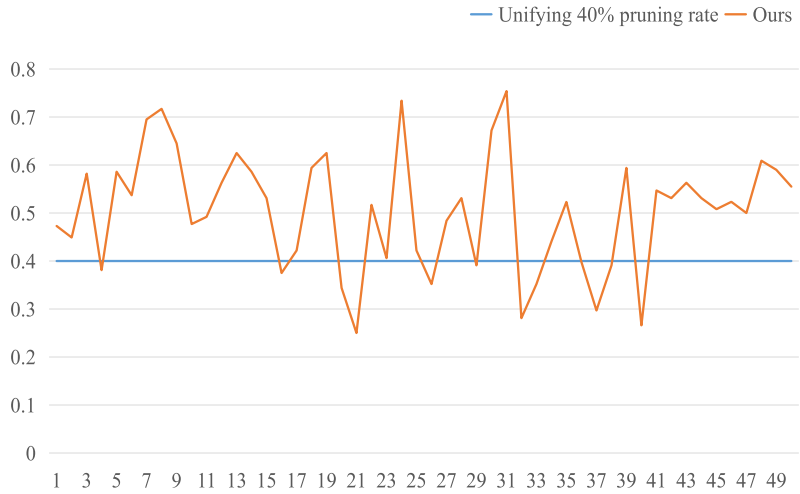


Fig. 8. Comparison of pruning strategies on the SSDD dataset.

pruning with a 20% channel number. Comparison experiment between the fine-tuning results and the final training results on the SSDD dataset.

D. Experiments on the SSDD Dataset

In this section, we conduct ablation experiments on the YOLOV3-tiny, YOLOV5, and YOLOX-tiny detectors using the SSDD dataset. We then compare the experimental outcomes with state-of-the-art SAR ship detection methods, contemporary target detection algorithms, and classical compression techniques.

1) Ablation Experiments on SSDD Datasets: To verify that SARGap can be applied to various SAR target detectors. In this section, the YOLOV3-tiny, YOLOv5, and YOLOX-tiny detectors were compressed and validated using the SSDD dataset. It mainly includes the following aspects.

1) In the latest target detection methods YOLOV3-tiny, YOLOV5s, and YOLOX-tiny, the SSDD dataset is used to train their detectors and obtain results;
2) Each group of network structures in the YOLOV3-tiny, YOLOV5s, and YOLOX-tiny detectors is pruned at a uniform 40% pruning rate, the pruned detectors are trained using the SSDD dataset, and the results are obtained.
3) SARGap is used to search the optimal pruning rate of each group of network structures in the YOLOV3-tiny, YOLOV5s, and YOLOX-tiny detectors, and the explored pruning rate was used to prune the detectors. Finally, the SSDD dataset is used to train the compressed detectors and obtain the results.

As shown in Table IV, three sets of experiments are conducted on YOLOV3-tiny, YOLOV5s, and YOLOX-tiny on the SSDD dataset. Three sets of experiments with three detectors on the SSDD dataset show that the SARGap method achieves good performance.

As shown in Fig. 8, the optimal pruning rate of each group of network structures searched by the SARGap method on the SSDD dataset based on the YOLOV5s detector is shown. The blue line represents the pruning rate of all network structures in the detector and is set to 40%, and the orange line represents the pruning rate of each group of network structures explored by the SARGap method.

2) Comparison Results with Classical Model Compression Methods on SSDD Datasets: We found some papers on model compression because the implementation methods of these papers are more complicated, and most of the code is for backbone pruning. In this article, the YOLOV5n backbone is first replaced with MobileNetV2, and then these methods are directly applied to the YOLOV5n backbone (MobileNetV2). The backbone is pruned, and the SSDD dataset is used to train the pruned model. In addition, when SARGap is used to automate the pruning of YOLOV5n, its backbone is also replaced

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TABLE V
COMPARISON OF RESULTS WITH CLASSICAL MODEL COMPRESSION METHODS

Method	Params(M)	FLOPs(G)	mAP50(%)
Base(YOLOV5n-MobileNetV2)	4.5	7.9	97.8
DHP[42]	2.4	3.7	97.7
Net-Slim[8]	2.3	5.1	97.7
CHipNet[43]	2.6	3.9	97.8
OTOV2[50]	2.1	3.3	97.8
MetaPruning[44]	4.2	2.8	97.8
Ours(YOLOV5n-MobileNetV2)	**1.1**	**2.3**	**97.8**

TABLE VI
COMPARISON OF RESULTS WITH THE LATEST SAR SHIP DETECTION METHODS

Method	Params(M)	FLOPs(G)	mAP50(%)
EFGRNet[51]	-	-	91.1
DCMSNN[52]	41.1	-	89.4
DAPN[53]	-	-	89.8
LPEDet[37]	5.7	18.4	97.4
FBR-Net[21]	32.5	-	94.1
AFSar[17]	-	9.9	97.7
CenterNet++[55]	-	-	95.1
MSSDNet[56]	-	-	95.6
SII-Net[57]	-	-	95.5
Ours(YOLOV5s)	**1.1**	**3.1**	**98.7**

TABLE VII
COMPARISON WITH THE LATEST TARGET DETECTION ALGORITHMS ON THE SSDD DATASET

	Method	Params(M)	FLOPs(G)	mAP50(%)
Anchor-free	CornerNet[58]	201.0	707.8	94.7
	CenterNet[59]	14.2	20.4	95.1
	FCOS[60]	61.0	78.6	95.3
	YOLOX-tiny[49]	5.0	15.1	96.2
	CenterNet++[55]	-	-	95.1
	FBR-Net[21]	32.5	-	94.1
	Ours(YOLOX-tiny)	**1.1**	**3.7**	96.0
Anchor-based	YOLOV3-tiny[47]	8.7	13.0	98.6
	FPN[62]	63.6	71.7	96.5
	Faster R-CNN[20]	41.1	91.4	96.4
	Cascade R-CNN[63]	69.2	119.1	96.8
	YOLOV5s[48]	7.0	15.9	98.9
	Ours(YOLOV5s)	**1.1**	**3.1**	98.7

by MobileNetV2, and the final result is compared with the above method. As shown in Table V, the SARGap method shows better performance than the other model compression methods.

3) Comparison Results With the Latest SAR Ship Detection Methods on the SSDD Dataset: As shown in Table VI, I also reviewed a large number of papers on SAR ship target detection, and since the implementation methods of these papers are rather cumbersome and most of the code is not open source, this article does not reproduce these algorithms but directly uses the experimental results in these papers. To compare with these algorithms and to prove the advanced nature of SARGap, this section uses SARGap to compress the YOLOV5s detector and uses the same dataset SSDD for training. The training parameters used in the training are also consistent with these methods. Since many of these tests are not considered from a lightweight point of view, many of these tests did not publish their numbers of parameters and calculations. This article selects the best results from their experimental results for comparison.

4) Comparison Results With the Latest Target Detection Algorithm on the SSDD Dataset: Table VII further demonstrates the advanced nature of SARGap's work. On the SSDD dataset, SARGap is used to automatically prune YOLOV5s

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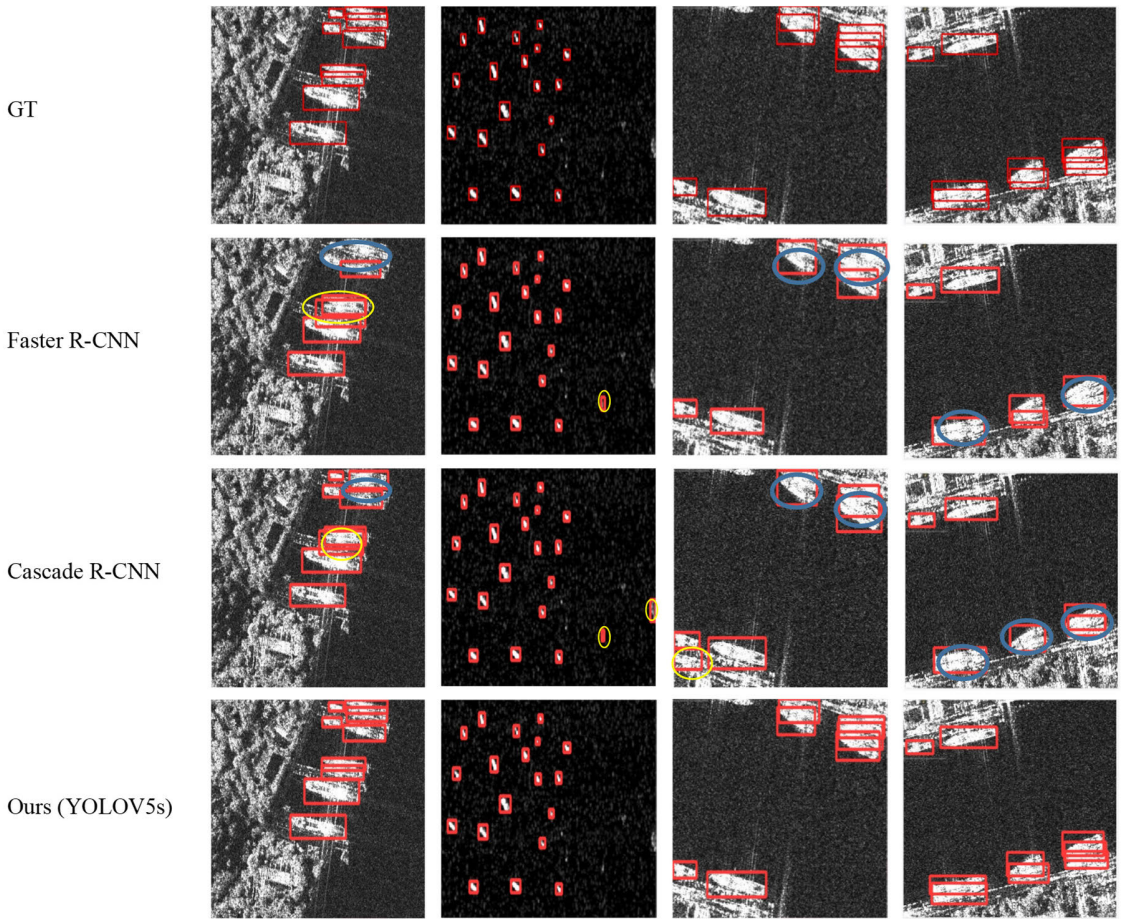


Fig. 9. Comparison of visual detection results with the latest target detection algorithms on the SSDD dataset.

TABLE VIII
RESULTS OF EXPERIMENTS CONDUCTED ON THE HRSID DATASET

	policy	Params(M)	FLOPs(G)	mAP50(%)
YOLOV3-tiny[47]	Base	8.7	20.3	90.6
	40% Pruned	3.1	7.5	89.6
	Ours	**2.6**	**6.7**	89.7
YOLOV5s[48]	Base	7.0	24.9	94.5
	40% Pruned	2.5	9.3	93.7
	Ours	**1.2**	**4.5**	93.5
YOLOX-tiny[49]	Base	5.0	23.7	88.7
	40% Pruned	1.8	8.8	88.0
	Ours	**1.3**	**6.6**	88.1

with anchor-based and YOLOX-tiny with anchor-free, and the performance of the prune detectors is compared with the latest target detection algorithms with anchor-based and anchor-free. After the automatic pruning of the YOLOV5s and YOLOX-tiny detectors, the model accuracy can perform well even with the minimum number of calculations and parameters. Compared with other detection methods, YOLOV5s can achieve good results in terms of model accuracy, number of parameters, and calculations after automatic pruning.

As shown in Fig. 9, a comparison of visual detection results on the SSDD dataset with the latest target detection algorithms is shown. The yellow circle box indicates the misdetected target, and the blue circle box indicates the missed target.

E. Experiments on the HRSID Dataset

In this section, a series of ablation experiments were performed on the YOLOV3-tiny, YOLOV5, and YOLOX-tiny detectors on the HRSID dataset, and the experimental results were compared with the detection effects of the latest target detection algorithms.

1) Ablation Experiments on the HRSID Dataset: As shown in Table VIII, this section conducts three sets of experiments on YOLOV3-tiny, YOLOV5s, and YOLOX-tiny on the HRSID dataset, similar to the experiments on the SSDD dataset. The first set of experiments used the HRSID dataset to directly train the three detectors to obtain experimental results. The second group of experiments uniformly sets the pruning rate

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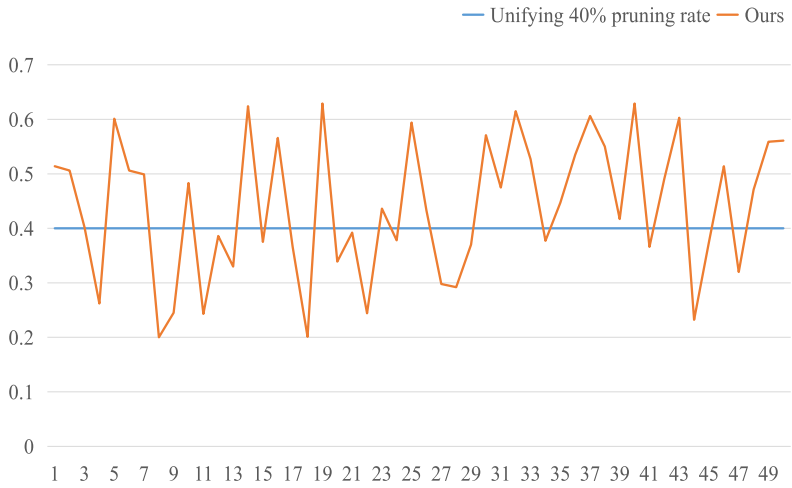


Fig. 10. Comparison of pruning strategies on the HRSID dataset.

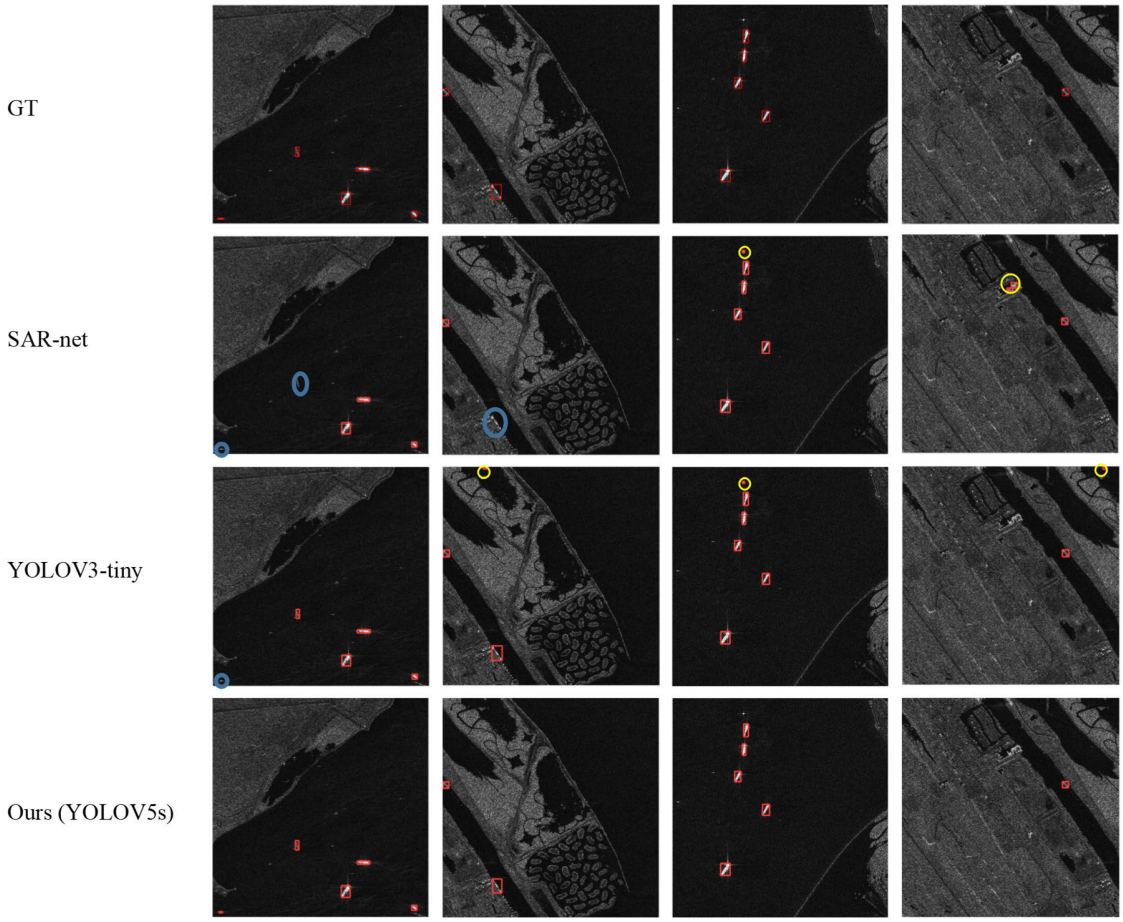


Fig. 11. Comparison of visual detection results with the latest target detection algorithms on the HRSID dataset.

of all network structures in the three detectors to 40% and then uses the HRSID dataset to train the detectors after pruning to obtain evaluation indicators. In the third group of experiments, the SARGap method is used to explore the pruning rate of the three detectors on the HRSID dataset, and then the pruning rate is used to prune the three detectors, and after pruning, the three detectors are trained by the HRSID dataset to obtain evaluation indicators.

As seen in the three groups of experiments shown in Table VIII, the SARGap method designed in this article performs well on the HRSID dataset. The comparison of pruning strategies is shown in Fig. 10, where the orange line shows the optimal pruning rate of each group of network structures in the YOLOV5s detector searched by SARGap on the HRSID dataset, and the blue line indicates that the pruning rate of all network structures in the detector is uniformly set to 40%.

2) Comparison Results With the Latest Target Detection Algorithm on the HRSID Dataset: As shown in Table IX, this article uses SARGap for automatic pruning on the HRSID dataset based on YOLOV5s with anchor-based and

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TABLE IX
COMPARISON WITH THE LATEST TARGET DETECTION ALGORITHMS ON THE HRSID DATASET

	Method	Params(M)	FLOPs(G)	mAP50(%)
Anchor-free	YOLOX-tiny[49]	5.0	23.7	88.7
	CenterNet2[54]	71.6	-	64.5
	LPEDet[37]	5.7	18.4	89.7
	Ours(YOLOX-tiny)	**1.3**	**6.6**	88.1
Anchor-based	Faster R-CNN[20]	41.5	91.4	74.0
	SAR-net[61]	42.6	104.2	84.7
	YOLOV3-tiny[47]	8.7	20.3	90.6
	YOLOV5s[48]	7.0	24.9	94.5
	Ours(YOLOV5s)	**1.2**	**4.5**	93.5

YOLOX-tiny with anchor-free and trained compressed detectors. The final training results are compared with the

latest target detection methods with anchor-based and anchor-free methods. The table shows that in the HRSID dataset, after the automatic pruning of models YOLOV5s and YOLOX-tiny, compared with other detection methods, the detector accuracy also performs well under the condition of the lowest number of calculations and parameters.

As shown in Fig. 11, the comparison between the visual detection results of the compressed YOLOV5s and the latest target detection algorithm on the HRSID dataset is shown. The yellow circle box indicates the misdetected target, and the blue circle box indicates the missed target.

V. CONCLUSION

When contemplating the use of a novel SAR target detector in conjunction with a new SAR image scene, there is no imperative need for extensive manual reimplementation of the algorithm. By employing the aforementioned target detector, the comprehensive automatic pruning algorithm, SARGap, can effectively attain an optimal equilibrium between target detector precision and computational efficiency. SARGap exhibits commendable versatility, rendering it suitable for integration with various detectors. The primary tasks involved in this context are as follows.

1) An automatic network structure grouping method, denoted as GSP, has been developed for arbitrary SAR target detectors. GSP possesses the capability to uniformly categorize all network structures associated with any given target detector. Initially, GSP generates a dependency graph based on automated analysis outcomes of the network structure within the SAR target detector. Subsequently, it organizes the pair-coupled network structures into groups in accordance with the dependency graph. This grouping strategy ensures that network structures within the same group undergo pruning with the same channel allocation, thereby mitigating the risk of rendering the target detector inoperative following the pruning process.
2) An automatic method for searching optimal pruning rates, referred to as APRS, has been devised. APRS operates entirely without human intervention and is proficient at autonomously seeking the optimal combination of pruning rates that strike an equilibrium between precision and computational efficiency.
3) A novel multi-objective optimization loss function, named MOOL, has been developed to serve as the objective function for APRS. MOOL is a versatile function capable of supervising precision, computational workload, and parameter count. It is designed to guide the model in an automated search for the optimal pruning rate combination that effectively balances accuracy and speed. This optimization process is performed in alignment with the specific requirements of the hardware platform's computational resources and precision criteria.

To verify the applicability of SARGap to any deep learning target detection network, the automatic compression of YOLOV3-tiny, YOLOV5s, and YOLOX-tiny detectors using SARGap on two large-scale SAR target detection datasets, SSDD and HRSID, performed well. Among them, on the YOLOV5s benchmark model, after automatic compression on the SSDD dataset with only 1.1M parameter number and 3.1G calculations, the mAP50 accuracy can reach 98.7%. The mAP50 accuracy can reach 93.5% by using only 1.2M parameters and 4.5G calculations on the HRSID dataset. This result is superior to the existing deep learning lightweight object detection methods and has good generality.

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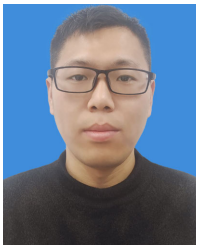
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Jingqian Yu was born in Anhui, China, in 1996. He is currently pursuing the master's degree with the Institute of Material Science and Information Technology, Anhui University, Hefei, China.

His research interests include remote sensing image classification and synthetic aperture radar (SAR) image target detection.



Jie Chen (Member, IEEE) received the Ph.D. degree in computer science and technology from the Hefei Institute of Physical Science, Chinese Academy of Sciences, University of Science and Technology of China, Hefei, China, in 2019.

He is currently an Associate Professor at Anhui University, Hefei, a Master Supervisor of the Information Materials and Intelligent Sensing Laboratory of Anhui Province, Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, and also a Post-Doctoral Fellow at the 38th Research Institute of China Electronics Technology Group Corporation, Hefei. He has presided over ten projects, including the National Natural Science Foundation Youth Project, Anhui Provincial Natural Science Foundation Youth Project, and the China Postdoctoral Science Foundation General Project. He has published dozens of papers in top journals/conferences, such as IEEE TRANSACTIONS ON INTELLIGENT TRANSPORTATION SYSTEMS and IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, AND RESEARCH. He holds more than ten invention patents that have been accepted and authorized. His research interests include deep learning, intelligent interpretation of remote sensing images, and autonomous driving.

Dr. Chen won the Silver Medal of Anhui Province in the First National Postdoctoral Innovation and Entrepreneurship Competition in 2021 and was awarded the honorary title of Excellent Postdoctoral Fellow.



Huiyao Wan was born in Henan, China, in 1995. He received the B.S. degree from the Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, School of Electronics and Information Engineering, Anhui University, Hefei, Anhui, China, in 2019, where he is currently pursuing the M.S. degree.

His research interests include remote sensing image classification and object detection for synthetic aperture radar (SAR) images.



Zheng Zhou was born in Chuzhou, China, in 1996. He received the B.S. degree from the Information Materials and Intelligent Sensing Laboratory of Anhui Province and the Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, School of Electronics and Information Engineering, Anhui University, Hefei, Anhui, China, in 2020, where he is currently pursuing the M.S. degree.

He is an Intern at the Earth Observation Research and Development Center, 38th Research Institute of China Electronics Technology Group Corporation, Hefei. He has published several high-level journal articles, such as IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, a top journal in the field of remote sensing. His research interests include few-shot object detection on synthetic aperture radar (SAR) images and object detection for multipolarization SAR images.



Yice Cao received the Ph.D. degree in the Remote Sensing Image Processing and Fusion Group, School of Electronic Engineering, Xidian University, Xi'an, China, in 2022.

She is currently a Lecturer at the Key Laboratory of Intelligent Computing and Signal Processing, Ministry of Education, Anhui University, Hefei, China. Her primary research focuses on the intelligent interpretation of remote sensing images and deep learning.



Zhixiang Huang (Senior Member, IEEE) was born in Anhui, China. He received the B.S. and Ph.D. degrees in computational electromagnetics from Anhui University (AHU), Hefei, China, in 2002 and 2007, respectively.

Since 2008, he has been a Full Professor with the School of Electronic Information and Engineering, AHU. From September 2010 to September 2011, he was a Visiting Scholar with Iowa State University, Ames, IA, USA. From August 2013 to October 2013, he was a Visiting Professor with the University of Hong Kong, Hong Kong. From February 2014 to February 2015, he was a Visiting Professor with the Beijing National Laboratory for Condensed Matter Physics, Institute of Physics, Chinese Academy of Sciences, Beijing, China. He has authored or coauthored one monograph and three book chapters for CRC Press, Boca Raton, FL, USA, and InTech Publishers, London, U.K. He has authored or coauthored 80 peer-reviewed journal papers included in the Web of Science Core Collection. His research interests include theoretical and computational research in electromagnetics and image, focusing on the multiphysics and interdisciplinary research, and fundamental and applied aspects in metamaterials and active metamaterials.

Dr. Huang is a member of OSA. He was a recipient of the Second Prize of Science and Technology from the Anhui Province Government, China, in 2015, the National Science Foundation for Outstanding Young Scholar of China, in 2017, and the Changjiang Distinguished Professorship of the Ministry of Education in 2020.

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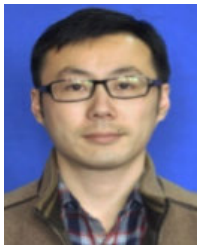
Yingsong Li (Senior Member, IEEE) received the B.S. degree in electrical and information engineering and the M.S. degree in electromagnetic field and microwave technology from Harbin Engineering University (HEU), Harbin, China, in 2006 and 2011, respectively, and the Ph.D. degree from both Kochi University of Technology (KUT), Kochi, Japan, and HEU, in 2014.

He was a Professor at Habirn Engineering University, from 2014 to 2022, a Visiting Scholar at the University of California, Davis, CA, USA, from March 2016 to March 2017, a Visiting Professor at the University of York, U.K., in 2018, and a Visiting Professor at Far Eastern Federal University (FEFU), Vladivostok, Russia, and KUT. He was a Post-Doctoral Fellow of Key Laboratory of Microwave Remote Sensing, Chinese Academy of Sciences, from 2016 to 2020. He has been a Full Professor at the School of Electronic and Information Engineering, Anhui University, Hefei, China, since March 2022. Now, he is a Visiting Professor at the School of Information of KUT, which he began in 2018. He has authored or coauthored approximately 300 journal and conference papers in various areas of electrical engineering. His research interests include remote sensing, underwater communications, signal processing, adaptive filters, metasurface designs, and microwave antennas.

Dr. Li is a fellow of Applied Computational Electromagnetics Society (ACES Fellow) and a Senior Member of Chinese Institute of Electronics (CIE). He was the General Co-Chair of ICEICT in 2020 and the General Chair of IEEE 9th International Conference on Computer Science and Network Technology (ICCSNT 2021) and ICCSNT in 2022. He also serves as the Session Chair or Organizer for many international and domestic conferences, including the WCNC, AP-S, and ACES. He is the TPC Co-Chair of the 2019 IEEE International Workshop on Electromagnetics (iWEM 2019-2020), 2019 IEEE 2nd International Conference on Electronic Information and Communication Technology (ICEICT 2019), 2019 International Applied Computational Electromagnetics Society (ACES) Symposium–China, 2019 Cross Strait Quad-regional Radio Science and Wireless Technology Conference (2019 CSQRWC), and the TPC Chair of ICEICT from 2021 to 2022. He served as an Area Editor of the AEÜ-International Journal of Electronics and Communications from 2017 to 2020, and he is an Associate Editor of IEEE ACCESS, *Applied Computational Electromagnetics Society Journal* (ACES journal), and *Alexandria Engineering Journal*. He acts as a reviewer of numerous IEEE, IET, Elsevier, and other international journals.



Bocai Wu was born in Xiaogan, Hubei, China. He is currently a researcher-level Senior Engineer at the 38th Research Institute of China Electronics Technology Group Corporation, Hefei, China. As the person in charge, he has presided over more than ten key radar projects and won nine provincial and ministerial-level scientific and technological progress awards in China. He has been granted ten invention patents and has published more than 20 high-level papers. His main research directions are artificial intelligence technology and application, reconnaissance and surveillance radar system design, and synthetic aperture radar (SAR) radar application.



Baidong Yao received the Ph.D. degree in optics from the University of Chinese Academy of Sciences, Hefei, China, in 2013.

He is currently a Researcher-Level Senior Engineer with the 38th Research Institute, China Electronics Technology Group Corporation, Hefei. His research interests include synthetic aperture radar (SAR) system design, SAR remote sensing image processing, and intelligence application technology.

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